

Drought recovery in plants triggers a cell-state-specific immune activation

Received: 24 February 2025

Accepted: 21 August 2025

Published online: 29 August 2025

 Check for updates

Natanella Illouz-Eliaz^{1,2,3}, Jingting Yu⁴, Joseph Swift^{1,2}, Kathryn Lande^{4,5}, Bruce Jow^{1,2,3}, Lia Partida-Garcia^{1,3}, Za Khai Tuang⁶, Travis A. Lee^{1,2,3}, Adi Yaaran⁷, Rosa Gomez-Castanon^{1,2}, William Owens², Chynna R. Bowman⁸, Emma Osgood², Joseph R. Nery^{1,2}, Tatsuya Nobori^{1,2,9}, Yotam Zait⁷, Saul Burdman⁶ & Joseph R. Ecker^{1,2,3} 

All organisms experience stress as an inevitable part of life, from single-celled microorganisms to complex multicellular beings. The ability to recover from stress is a fundamental trait that determines the overall resilience of an organism, yet stress recovery is understudied. To investigate how plants recover from drought, we examine a fine-scale time series of RNA sequencing starting 15 min after rehydration following moderate drought. We reveal that drought recovery is a rapid process involving the activation of thousands of recovery-specific genes. To capture these rapid recovery responses in different *Arabidopsis thaliana* (*A. thaliana*) leaf cell types, we perform a single-nucleus transcriptome analysis at the onset of drought recovery, identifying a cell type-specific transcriptional state developing independently across cell types. To further validate the cell-type specific transcriptional changes observed during drought recovery, we employ spatial transcriptomics using multiplexed error-robust fluorescence in situ hybridization (MERFISH), revealing anatomical localization of recovery-induced gene expression programs across *Arabidopsis* leaf tissues. Furthermore, we reveal a recovery-induced activation of the immune system that occurs autonomously, and which enhances pathogen resistance in vivo in *A. thaliana*, wild tomato (*Solanum pennellii*) and domesticated tomato (*Solanum lycopersicum* cv. M82). Since rehydration promotes microbial proliferation and thereby increases the risk of infection, the activation of drought recovery-induced immunity may be crucial for plant survival in natural environments. These findings indicate that drought recovery coincides with a preventive defense response, unraveling the complex regulatory mechanisms that facilitate stress recovery in different plant cell types.

In most plants, extended periods of water deficit result in reduced growth, premature flowering, flower abortion, fruit abscission, and, ultimately, decreased yield^{1,2}. Plant responses to drought have therefore been studied extensively as part of efforts to develop strategies or genetic manipulations that could mitigate the economic and agricultural consequences of future droughts.

Previous studies have identified and functionally characterized numerous genes that respond to water deficit, including various transcription factors (TFs) that regulate plant drought responses^{2–4}. In general, drought stress in plants induces dramatic changes in the transcriptional landscape^{5,6}. For example, the rapid up-regulation of genes involved in osmolyte accumulation enables water retention by

A full list of affiliations appears at the end of the paper.  e-mail: ecker@salk.edu